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import nltk
from nltk.corpus import brown
from collections import defaultdict, Counter

nltk.download('brown')

[nltk_data] Downloading package brown to /root/nltk_data...
[nltk_data]   Unzipping corpora/brown.zip.

True

sentences = brown.sents()

# Tokenization + Lowercase
tokens = [word.lower() for sent in sentences for word in sent]

print("Total Tokens:", len(tokens))
print("Sample Tokens:", tokens[:20])

Total Tokens: 1161192
Sample Tokens: ['the', 'fulton', 'county', 'grand', 'jury', 'said',
'friday', 'an', 'investigation', 'of', "atlanta's", 'recent',
'primary', 'election', 'produced', '``', 'no', 'evidence', "'",
'that']

def build_ngram(tokens, n):
    model = defaultdict(Counter)
    for i in range(len(tokens) - n + 1):
        context = tuple(tokens[i:i+n-1])
        word = tokens[i+n-1]
        model[context][word] += 1
    return model

bigram_model = build_ngram(tokens, 2)
trigram_model = build_ngram(tokens, 3)

print("Bigram contexts:", len(bigram_model))
print("Trigram contexts:", len(trigram_model))

Bigram contexts: 49815
Trigram contexts: 436002

def predict_next(context, model):
    context = tuple(context.lower().split())
    if context not in model:
        return "Context not found."

    total = sum(model[context].values())
    predictions = [(word, round(count/total, 4))
                    for word, count in model[context].most_common(5)]
    return predictions

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print("Bigram Prediction for 'government':")
print(predict_next("government", bigram_model))

print("\nTrigram Prediction for 'the government':")
print(predict_next("the government", trigram_model))

print("\nTrigram Prediction for 'in the':")
print(predict_next("in the", trigram_model))

Bigram Prediction for 'government':
[('of', 0.1364), ('.', 0.1077), (',', 0.0813), ('and', 0.0359),
 ('was', 0.0287)]

Trigram Prediction for 'the government':
[('of', 0.3664), (',', 0.0534), ('.', 0.0382), ('was', 0.0382),
 ('and', 0.0305)]

Trigram Prediction for 'in the':
[('world', 0.0149), ('first', 0.0146), ('united', 0.0123), ('same',
0.0116), ('past', 0.0098)]
```